Automatic supervision of big Earth data to increment the reliability of land cover datasets for post-disaster recovery monitoring

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Abstract:	Analyzing post-disaster resettlement in the periphery of urban areas is essential to avoid rural fragmentation. With the use of remote sensing satellite images, land change effects of resettlement can be quantified to inform pre-disaster recovery plans. However, change detection is governed by the reliability of training samples and the efficiency of learning models. Therefore, we formulated an automatic label supervision model to secure reliable land cover samples from multisource remote sensing data and predict accurate land cover maps, aiming to evaluate land change interactions prompted by post-disaster resettlement sites. Pre-existing land cover samples were photo-inspected and then vetted against fused-canopy heights to alleviate interpretation biases. Then, a set of refined labels were held as reference samples while the prevailing quality samples derived statistical amplitude metrics to automatically supervise the feature space and structural height of candidate training samples. A classifier was then optimized to predict Sentinel-2 satellite images over Java, Indonesia. We observed 16.4% forest loss, 18.5% cropland growth, and 15.5% built-up growth as a direct consequence of urban resettlement after the 2010 Mt. Merapi eruptions. With 32% bias-reduced reference samples and 56% error-removed training data, 82%-78% overall accuracies were achieved on the first iteration. Codes are available at: https://github.com/martingarciafry/GCC_model.
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remote sensing community.	Wen Liu, Ph.D. Assistant Professor, Chiba University wen.liu@chiba-u.jp Professor Wen Liu has published research articles focusing on damage mapping, emergency response, and early recovery scenarios. Her expertise covers multimodal and multitemporal Earth observation as proven by her unique contributions to the scientific community.
	Cristina Gomez, Ph.D. Post-Doctoral Researcher, University of Valladolid Higher Technical School of Agricultural Engineering of Palencia cgomez@uva.es Professor Cristina Gomez has conducted extensive research on land use and land cover time-series analyses. Her research has portrayed fascinating outcomes regarding forest and carbon stock monitoring, disturbance recovery, and

	land change with environmental implications. Moreover, the work carried out by her colleagues are equally impressive in the same areas of expertise.
urban risk-related topics. Professor Dell'Acqua has extensive experience in damage mapping and multisource remote sensing data applications.	Fabio Dell'Acqua, Ph.D. Professor, University of Pavia fabio.dellacqua@unipv.it Professor Fabio Dell'Acqua is a senior member of IIEE and a radar/optical data fusion specialist in agricultural and urban-risk related topics. Prof. Dell'Acqua has extensive experience in damage mapping and remote sensing data applications.
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September 2023

Jonathan Li, Ph.D.
Editor In Chief

International Journal of Applied Earth Observation and Geoinformation

Dear professor Li:

We would like to submit an article for publication in the International Journal of Applied Earth Observation and Geoinformation titled: “Automatic supervision of land cover labels to increment the reliability of training datasets for post-disaster recovery monitoring”. This paper is coauthored by professors Osamu Murao, Bruno Adriano, and Shunichi Koshimura.

In this study, we formulated a simple and automatic label supervision model to secure reliable land cover samples from multisource remote sensing data and predict accurate land cover maps, aiming to evaluate land change interactions prompted by post-disaster resettlement sites. We believe that our study makes a significant contribution to the literature for two reasons:

1. Multitemporal statistical metrics have been used to source land cover labels via trained models, or to classify images directly, but they have not been retained from photo-inspected samples vetted against fused-canopy height measures to automatically supervise the feature space and vertical height of candidate training samples.

2. In addition, the reliability of ‘bias-reduced’ reference samples (highly vetted samples) was assessed to measure the size of error inherent to simple photo-interpretations. We found that bias-reduced samples eliminated 32% of the mean variance in user’s accuracies, thus increasing the reliability of reported accuracies.

We also evaluated rural-affected areas after the 2010 Mt. Merapi eruptions in the Cangkringan district of Yogyakarta, Java, Indonesia, to clarify whether post-disaster resettlement sites avail land cover degradation. We analyzed 2016-2021 time-series maps in a high-spatial resolution domain to separate peri-urban growth and quantify land change implications of resettlement. Given the recurrence of natural disasters in Java, we also believe this study contributes with monitoring methods that facilitate the quantification of natural ecosystems exposed to multiple sources of disturbance (e.g., disaster damage, disaster recovery, urban expansion, informal growth).

Further, this paper opens the accessibility of remotely sensed image classification tasks to people with no extensive domain knowledge by vetting land cover labels against very-high resolution satellite images and using canopy height datasets to alleviate interpretation biases. This allows users to create and maintain regionally consistent and highly reliable training datasets for land cover classifications on-demand. We believe this might be of particular interest to the readership of your journal because being able to accumulate large quantities of high-quality labels is a widely known practical challenge. Moreover, to be able to apply the proposed method with no limits to spatial resolutions, sensors, or temporal acquisitions in regional extent might be considered of great utility. Lastly, the paper addresses, with social and environmental concern, the recovery of peri-urban areas where deforestation is likely and preservation is necessary.

This manuscript has not been published or presented elsewhere in part or in entirety and is not under consideration by another journal. We have read and understood your journal’s policies, and we believe that neither the manuscript nor the study violates any of these. The corresponding author will take full responsibility of notifying coauthors in a timely manner of any editorial decisions, reviews, or content revisions in response to further editorial review. All authors have read and approved the manuscript and there are no conflicts of interest to declare.

Thank you for your time.

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Automatic supervision of land cover labels to increment the reliability of training datasets for post-disaster recovery monitoring

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Abstract: Analyzing post-disaster resettlement in the periphery of urban areas is essential to avoid rural fragmentation. With the use of remote sensing satellite images, land change effects of resettlement can be quantified to inform pre-disaster recovery plans. However, change detection is governed by the reliability of training samples and the efficiency of learning models. Therefore, we formulated an automatic label supervision model to secure reliable land cover samples from multisource remote sensing data and predict accurate land cover maps, aiming to evaluate land change interactions prompted by post-disaster resettlement sites. Pre-existing land cover samples were photo-inspected and then vetted against fused-canopy heights to alleviate interpretation biases. Then, a set of refined labels were held as reference samples while the prevailing quality samples derived statistical amplitude metrics to automatically supervise the feature space and structural height of candidate training samples. A classifier was then optimized to predict Sentinel-2 satellite images over Java, Indonesia. We observed 16.4% forest loss, 18.5% cropland growth, and 15.5% built-up growth as a direct consequence of urban resettlement after the 2010 Mt. Merapi eruptions. With 32% bias-reduced reference samples and 56% error-removed training data, 82%-78% overall accuracies were achieved on the first iteration. Codes are available at: https://github.com/martingarciafry/GCC_model.

Keywords: 2010 Mt. Merapi eruptions, Java, urban recovery, disaster resilience, multisource remote sensing.

1. Introduction

Natural disaster events have surged during the past five decades increasing the vulnerability of populated areas at risk (MacManus et al., 2021; Pavlinovic, 2021). Ready-to-use, updated, and accurate land cover datasets are a necessary source of information to determine the extent of damage, response, and recovery after disaster events. The priority of a response stage is to distribute relief and rescue operations, while recovery stages prioritize progress beyond the reconstruction of buildings and infrastructure (Contreras et al., 2016). A considerable amount of money from donors and governments is destined to finance the recovery process, but allocation and management are crucial to overcoming expectations of development endorsed by the Sendai Framework for Disaster Risk Reduction 2015-2030 (UNISDR, 2015).

Monitoring the urban fringe of developing cities helps to understand the dynamics of risk associated with informal growth patterns (Fekete, 2022), climate change (Gibson et al., 2016), and inconsistent planning schemes (Wijaya, 2018). Multitemporal time-series analyses of remote sensing data have been increasingly applied in these areas (Li et al., 2018; Shaw and Das, 2018). A number of studies have focused on pre- and post-disaster damage assessments using optical remote sensing (Koshimura et al., 2020), SAR images (Moya et al., 2020), UAV photogrammetry (Calantropio et al., 2021), and multimodal/temporal datasets (Adriano et al., 2021). However, very few studies examine the effects of resettlement and recovery on socioeconomic (land use) and environmental (land cover) change, often aiding informal land transactions (Ingwani, 2019). Further, global targets such as biodiversity loss and food security have been linked to land use extent affecting carbon sources and partial land cover degradation (Winkler et al., 2021). This makes urban growth a necessary implication of recovery plans, and quantifying its dynamics, a critical task in supporting these debates.

In terms of open data, moderate-resolution satellite images together with airborne and spaceborne lidar-derived height measures, and surface characterization, three-dimensional accounting for variability estimation has become possible within any given area (Wulder et al., 2020). Two spaceborne lidar instruments were launched in 2018, (1) the Global Ecosystem Dynamics Investigation (GEDI) on the International Space Station (ISS; Dubayah et al., 2020), and (2) the Advanced Topographic Laser Altimeter System (ATLAS) on-board the Ice, Cloud, and land Elevation Satellite-2 (ICESat-2; Markus et al., 2017). These instruments have brought innovations, and challenges, to the observation of dynamic ecosystems (Mulverhill et al., 2022).

In practical remote sensing applications, abrupt changes on the pixel level (e.g., change caused by human activity or natural disturbances) are difficult to estimate compared to synchronized change (e.g., vegetation growth and change in soil moisture). This is mainly due to the size of change in fractions of a pixel and the spectral magnitude of changes (Xu et al., 2022; Zhu et al., 2022). The predictive uncertainty of algorithms can also stream inconsistent time-series data (Zhao et al., 2019). This can be attributed to machine and deep learning models in the computer vision field suffering from insufficient and inefficient labels (Huang et al., 2015). Manual labeling is hampered by the time-consuming process to supervise all samples, while domain expertise is limited to particular sites (Li et al., 2022). Another big challenge for remote sensing data learners is label noise (Burke et al., 2021). Given the variety and complexity of applications, datasets with perfect annotations rarely exist (Zhou et al., 2023).

Researchers have explored various ways to overcome these issues using unsupervised, semi-supervised, and self-supervised learning models. Transfer learning and domain adaptation networks are unsupervised methods where a model is pre-trained on a source image to be applied on a target image elsewhere (Saha et al., 2022). The accuracy of these networks depends on the difference between source and target data. Therefore, weakly-supervised models offer an alternative solution entailing: (i) incomplete supervision, where a subset of training labels are available; (ii) inexact supervision, where coarse-grained labels are given; and, (iii) inaccurate supervision, where the given labels are not always true on the ground (Yue et al., 2022). Active learning is another method that requires a small number of labeled samples to draw instances according to some measures (Desai and Ghose, 2022). Likewise, self-supervised models use unlabeled data to identify features and labeled samples to adapt the model for downstream tasks (Wang et al., 2022). However, the reliability of labels can be contested due to interpretation biases. Therefore, how can we obtain accurately-labeled samples to increase the reliability of model predictions for pixel-wise change detections?

Motivated by the complexity, computing requirements, and labeling bias of existing models, we formulated a simple and automatic label supervision model to secure reliable land cover samples from multisource remote sensing data and predict accurate land cover maps, aiming to evaluate land change interactions prompted by post-disaster resettlement sites. The proposed model utilizes a representative lidar acquisition to query pre-existing land cover samples and rectifies those labels with very-high resolution (VHR) satellite imagery. Then, ICESat-2 canopy heights are fused with GEDI-derived forest heights to vet the structural height of interpreted classes. Bias-reduced samples are held as reference or metric units to draw static and multitemporal amplitude metrics per class. These metrics converge in a canopy height fusion system to automatically supervise the three-dimensional space of candidate training samples. This model allows users to maintain regionally-consistent and incrementally-reliable training datasets for land cover classifications on-demand.

The contribution of this study is two-fold: First, multitemporal metrics have been used to source labels via trained models (e.g., Tong et al., 2020), or to classify images directly, but have not been retained from bias-reduced samples to parse a canopy height vetting code and automatically supervise the feature space and vertical height of training samples. Second, bias-reduced reference samples were assessed to measure the size of error inherent to interpretation bias. We found that bias-reduced samples eliminate 32% of the mean variance, thus increasing the reliability of reported accuracies (Foody, 2010). These potential novelties overcome the difficulties in sourcing high-quality labels for land cover mapping.

The following section introduces our study area with a brief methodological description. Section 3 summarizes the data employed and Section 4 details the proposed model. Section 5 evaluates target areas with results discussed in Section 6, drawing concluding remarks.

2. Study area and methodology

2.1 The Cangkringan district, Yogyakarta, Java, Indonesia

From October 26 to November 30, 2010, the Merapi volcano eruptions displaced thousands of people from their homes causing massive damage to land and property (Kuncoro et al., 2012). The Cangkringan district, located in Yogyakarta, Java, Indonesia, was severely affected driving the relocation of 2,608 households onto 23 urban resettlement sites (Singgih and Asano, 2019).

Indonesia, situated at the intersection of three active tectonic plates (the Eurasian, Australian, and Pacific plates), boasts 129 active volcanos with the largest population globally (8.6 M) living in close range (10 km) to a volcano (Bachri et al., 2015). Java itself holds 33 active volcanos with Mt. Merapi being the most active stratovolcano in Indonesia due to its recurrent eruptions (Rani et al., 2021). Although living with volcanoes can be extremely dangerous, this region is part of the Java-Bali tropical rainforest ecoregion, located in the Indo-Malay ecozone with abundant natural resources and fertile agrarian land (Olson et al., 2001). Much of this land is explicitly fertile due to the sedimentation deposits (i.e., volcanic ash and sand) (Utami et al., 2018). Local farmers in the Cangkringan district choose to capitalize on increased yield after eruptions, but they are also exposed to Yogyakarta’s spillover effects in unprepared, informal, and high-risk areas (Rahajeng et al., 2022). These peri-urban areas are transitional spaces where neither decent urban services nor rural resource provision exist. The reader is directed to Hutchings et al. (2022) for a complete breakdown on this perspective.

Given the extent of vulnerable areas in Java, subject to multiple types of hazards (e.g., flash floods, earthquakes, tsunamis, volcanos, etc.), a 10 x 10 km grid over >13.4 Mha of vegetation ecotones was used to partition the territory in a WGS84 geodetic datum (Fig. 1). The resulting 1,392 tiles (111,100 pixels per tile) were overlaid the tropical and subtropical moist broadleaf ecosystems of Java to predict target areas in a high spatial resolution domain.

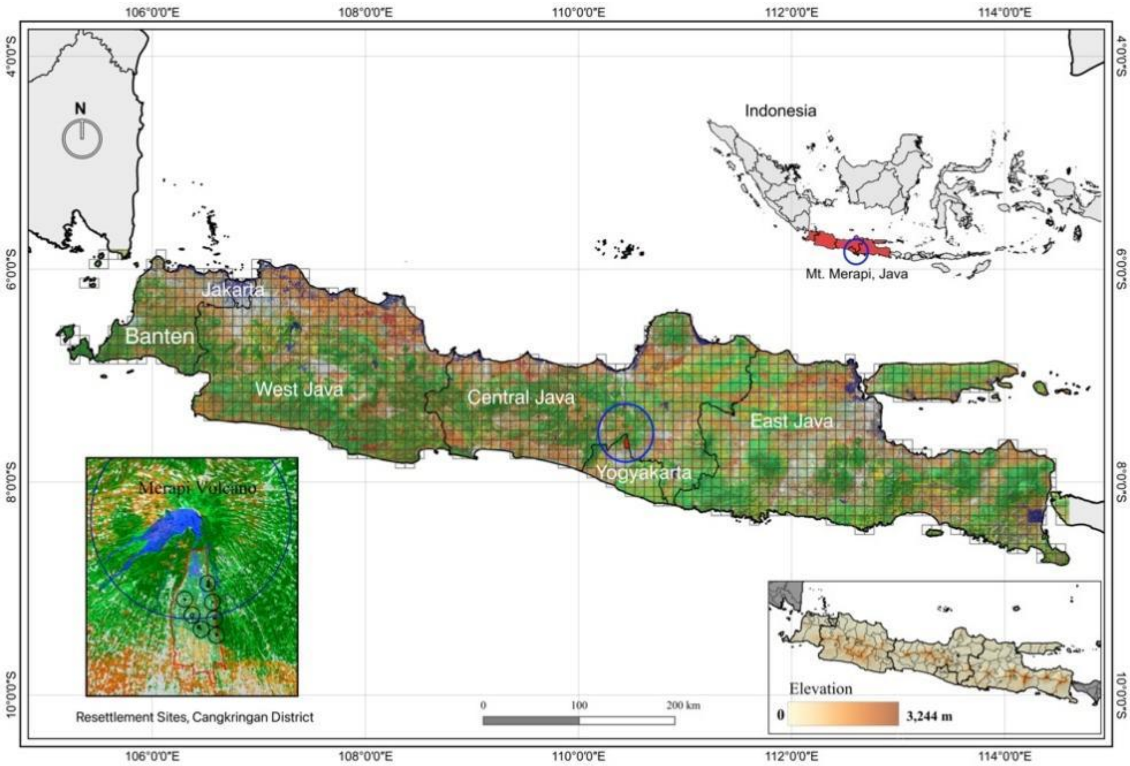

Fig. 1. Java's ecosystem overlaid by a classification grid of 10 x 10 km tiles. The Merapi volcano is located in the Special Region of Yogyakarta with the Cangkringan district directly below extending over peri-urban areas. The left inset box shows seven resettlement areas for evaluation near the 10 km hazard-prone radius and the right inset box shows elevation data. **(2 columns) (B/W)**

2.2 Methodology

The model termed ground cover change (GCC) accumulates high-quality training and reference data to predict satellite images with no limits to spatial resolutions, sensors, nor temporal acquisitions in regional extent. Here, we offer an overview of the three data processing pipelines, illustrated in Fig. 2 and detailed in sub-section 4.1 through sub-section 4.3. These steps do not require heavy computational resources; they ran on a standard PC with 2.4 GHz and a Quad-Core Intel Core i5 processor with 8 GB of RAM.

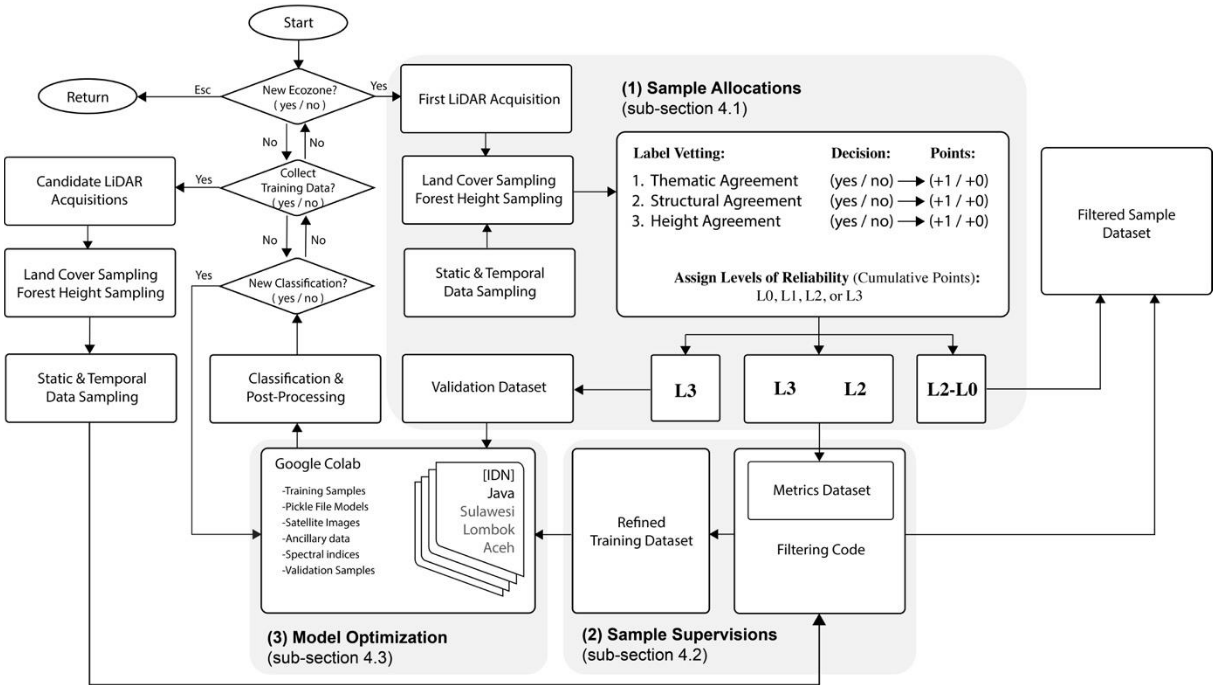

Fig. 2. Model architecture. **(2 columns) (B/W)**

(1) Sample allocations: A dense ICESat-2 ground-track acquisition was selected to sample land cover ([Dataset 1] Hansen et al., 2022) and forest height ([Dataset 2] Potapov et al., 2021) pixel landings. These samples were inspected to rectify mislabeled units and then vetted against fused values of forest structure to redeem interpretation errors based on the following agreed (1 point) or disagreed (0 points) binary decisions: First, thematic agreements were given to samples having ‘true’ labels using VHR satellite imagery; second, structural agreements were attributed to samples with ICESat-2 and forest height value correlations; and third, height agreements were given to samples measured against discrete forest height thresholds. These thresholds represent the structural height of vegetated/non-vegetated classes following a variant of FAO’s Land Cover Classification System (FAO, 2012). Cumulative scores rendered a hierarchical set of samples sorted by their level (L) of reliability enabling bias-reduced reference (L3) and metric (L3 and L2) samples to be held with indices in Table 1.

(2) Sample supervisions: Multinomial regressions were subsequently performed over discrete metric sample datasets. The selection of representative class features followed traditional remote sensing literature, but only those with statistically significant p-values (≥90%) were retrieved. Then, we extrapolated their statistical metrics to automatically detect the feature space of reliable candidate training samples.

(3) Model optimization: The training data was assessed through feature correlations and overfitting trials to optimize a classifier in Google Colab; a user-friendly platform for machine learning (Bisong, 2019). Uncorrelated features with the best relative contribution to land use classes were extracted using SHAP (Shapley Additive exPlanations) values (Lundberg and Lee, 2017). We found ten candidate feature-parameter combinations and discriminated the predictive loss in error. Then, preliminary accuracies revealed the utility of bias-reduced reference samples followed by post-processing and post-informed change mapping.

Type	*Resolution*	*Acronym/Indices*	*Dataset*	*Description/Formula*	*Reference*
Lidar	11 m footprint	ICESat-2 (Canopy heights)	ATL08_20190904163339	First ground track acquisition	(Neuenschwander and Pitts, 2019)
	11 m footprint	ICESat-2 (Canopy heights)	ATL08_20190904163339 ATL08_20190429105420 ATL08_20190523214429 ATL08_20190626080618 ATL08_20190713184723 ATL08_20190802180554 ATL08_20190621202031 ATL08_20190724184808	Training acquisitions	
	11 m footprint	ICESat-2 (Canopy heights)	ATL08_20190605205350 ATL08_20190610183939	Post-informed acquisitions	
Land Cover	~30 m	GLCLU	Global Land Cover Land Use, 2019	Global land cover product	(Hansen et al., 2022)
	~30 m	Forest Heights	Global Forest Canopy Heights, 2019	GEDI-derived forest heights	(Potapov et at., 2021)
DEM	~90 m	MERIT-DEM	Multi-Error-Removed Improved-Terrain DEM	Digital Elevation Model	(Yamazaki et al., 2017)
	~90 m	Slope	DEM-Derived Slope	Ancillary Dataset	
	~90 m	TSRI	Transformed Solar Radiation Index	Ancillary Dataset	(Roberts and Cooper, 1989)
	~90 m	TWI	Geomorpho90m- Topographic Wetness Index	Ancillary Dataset	(Amatulli et al., 2020)
Sentinel-2 Spectral Indices	~10m, 20m	TCT	Tasseled Cap Transformation	Coefficients for TCG and TCW	(Shi and Xu, 2020)
	~10m, 20m	TCG	Tasseled Cap Greenness	-0.3599*B -0.3533*G-0.4734*R+0.6633*NIR-0.0087*SWIR1-0.2856*SWIR2	(Kauth and Thomas, 1976)
	~10m, 20m	TCW	Tasseled Cap Wetness	0.2578*B+0.2305*G+0.0883*R+0.1071*NIR-0.7611*SWIR1-0.5308*SWIR2	
	~10m	NDVI	Normalized Difference Vegetation Index	NIR-R/NIR+R	(Tucker, 1979)
	~10m	GNDVI	Green Normalized Difference Vegetation Index	NIR-G/NIR+G	(Gitelson et al., 1998)
	~10m	NDWI 1	Normalized Difference Water Index 1	NIR-SWIR1/NIR+SWIR1	(Gao, 1996)
	~20m	NDWI 2	Normalized Difference Water Index 2	G-NIR/G+NIR	(McFeeters, 1996)
	~10m, 20m	RS2	Normalized Difference Flooding Index	R-SWIR2/R+SWIR2	(Boschetti et al., 2014)
	~20m	NS2	NIR-SWIR2	NIR-SWIR2/NIR+SWIR2	(Potapov et al., 2021)

~10m	SAVI	Soil-Adjusted Vegetation Index	(NIR-R/NIR+R+0.5)*1.5	(Huete, 1988)
~10m, 20m	IRECI	Inverted Red-Edge Chlorophyll Index	(B7-R / (B5 / B6))	(Frampton et al., 2013)
~10m	BN	Vegetation index	B-NIR/B+NIR	(Potapov et al., 2020)
~20m	S12	Vegetation index	SWIR1-SWIR2/SWIR1+SWIR2	(Potapov et al., 2020)

Table 1. Indices and datasets used for metric characterizations and image classifications. **(2 columns)**

3. Data

Mapping products, shape files, and datasets supporting findings of this study can be found in Mendeley Data ([Dataset 3] Garcia-Fry, 2023).

3.1 Lidar

Lidar is a unique source of comprehensive and explanatory data for forest inventories and ice detection with a range of attributes (e.g., height, volume, structure, biomass, and canopy cover) (Mulverhill et al., 2022). In a study by Liu et al. (2021), ICESat-2's Land and Vegetation Height (ATL08) data product was found to present fewer noise and almost same accuracy to GEDI data for nighttime and strong beam acquisitions. However, the best estimate of vegetation structures is reasoned by the fusion of high-powered laser systems such as GEDI and ICESat-2 (Neunschwander et al., 2019). ICESat-2 (ATL08, RH98) were selected rather than the maximum canopy photons owing to the signal-to-noise uncertainty at the top of canopy (Neuenschwander et al., 2021). The 98^{th} percentile of canopy heights in 20 m segments (h_canopy_20) and mean heights in 100 m segments (h_mean_canopy) were retrieved during evening and nighttime acquisitions between April 29 and September 04, 2019.

3.2 Land cover

The Global Land Cover and Land Use map for 2019 (Hansen et al., 2022) was selected to improve upon land use accuracies. This Landsat-derived mapping product includes maximum vegetation cover (percent per pixel), class heights, and ecozones in an overall climate domain classification system (FAO, 2012). The maps were generated in a consistent framework with multitemporal and statistical metrics for classification. Statistical measures interpret phenological and morphological time-sequencing factors, or attributes of a given class. They retain information without regard to time or place, provided they represent a statistically valid representation of an area (Song et al., 2001). While many studies derive training data from pre-existing land cover maps (Hermosilla et al., 2022; Inglada et al., 2017; Wessels et al., 2016), spurious samples can propagate errors to the classification. So, vetting and rectifying these samples is essential to the classifier's performance, and should be accomplished through automated or simple methods.

3.3 Digital elevation model

The Multi-Error-Removed Improved-Terrain Digital Elevation Model (MERIT-DEM) is a general-purpose elevation product at 3 arc-seconds resolution over land areas between 90N-60S (Yamazaki et al., 2017). This product was selected given the multi-source DEM's used to remove errors: NASA's SRTM3, JAXA AW3D-30 m DEM, Viewfinder Panorama's DEM, including ancillary ICESat/GLAS data products, a 2000-2010 global forest height product (Hansen et al., 2013), NASA's global forest height product, and the Global 3 arc-second Water Body Map (Yamazaki et al., 2015). Land areas were mapped with significant improvement over flat regions, river networks, and hill-valley structures. In this study, we use MERIT-DEM to compute topographical and hydrological features such as slope (in degrees), TSRI–a transformed measure of solar radiation aspect (Roberts and Cooper, 1989), and Geomorpho90m–a MERIT-DEM-derived wetness index (Amatulli et al., 2020).

3.4 Sentinel-2

Sentinel-2, Level-1C images with provided in Top-of-Atmosphere (TOA) reflectance's were collected from the Google Earth Engine (GEE) cloud platform (Gorelick et al., 2017). The

Sentinel-2 archives offer satellite image collections from 2015-onwards with moderate resolutions (~10, 20, and 60 m) useful to land cover change detection due to the scale of anthropogenic factors (Townshend et al., 2012). Our first concerns were directed towards near-complete coverage from cloud cover as of 2016. The median value of image collections between April and August (i.e., the local vegetation cycle) were retrieved with <20% cloud cover over Java. Pre-processing from TOA to Bottom-of-Atmosphere (BOA) reflectance's followed with the SIAC module on GEE (Yin et al., 2022). 2016-2021 annual images were mosaicked into seamless composites west (10°S, 105°E–66,918,389 pixels) and east (10°S, 110°E–79,826,814 pixels). Intra-annual phenological stages, which can be difficult to obtain even for cloud-optimized images due to the atmospheric back-scattering propagation derived from stratospheric aerosols, cirrus clouds, and other elements (e.g., mixed-phase clouds, water, light, etc.) (Shelestov et al., 2017), were sought for the Cangkringan district in 2017, 2018, 2020, and 2021. All images were projected in WGS 84 to match the ATL08 product and wavelengths were resampled to a ~30 m target resolution with the nearest neighbor algorithm.

4. Ground Cover Change model

The GCC model adopts seven strata divided into vegetated and non-vegetated land cover and land use classes. Vegetated classes consist of dense short vegetation (≤5 m, stable clusters and open sparse vegetation), open tree cover (5 m ≤ h <12 m, stable open trees), dense tree cover (≥12 m, clustered trees), and rice paddies/wetlands (<12 m, open trees and water presence). Non-vegetated classes include built-up (<8 m, built structures) and water (0 m, inland bodies of water). Cropland is regarded as cultivated or uncultivated areas (≤5 m), further stratified into growing peak, senescence, and uncultivated sub-classes.

4.1 Sample allocations

The first ICESat-2 lidar points were acquired in 20 m segments along-track for a wide and heterogenous transect of Java. We sampled the land cover product and used a morphological erosion window to disregard points landing ≥45 m away from inter-class perimeters. This accounts for class variability while avoiding mixed pixels in areas with thematic homogeneity (Wulder et al., 2020). The prevailing points were vetted against quality control mechanisms using a three-point-based scoring system to arrange the pool of samples by levels of reliability (Fig. 3). The first mechanism queries land cover landings and seeks thematic agreement based on VHR satellite images in Google Earth. Suppose a lidar footprint lands over a fraction of a pixel p with label y_n and area A_p where two mixed classes, i and j, belong to a set of n classes:

$$T_p = \{(i,j) \mid j \in J_p, \forall\, A_i < A_j\}; \quad (1)$$

$$y_j = \{j \mid j \in J_p, \textstyle\sum_i (A_i) < A_p/2\}; \quad (2)$$

$$y_i = \{i \mid i \in I_p, \textstyle\sum_j (A_j) < A_p/2\}; \quad (3)$$

$$Agreement_p = \{y_n \mid y_n \in Y_p, \forall\, A_n \geq A_p/2\}. \quad (4)$$

Equations 1-4 define whether ≥50% of a pixel is in agreement with the queried label. The second mechanism apportions a structural agreement to each sample k with forest heights, F_h^k, correlated with canopy heights, C_h^k, where $F_h^k = C_h^k \pm 6\ m$. A third mechanism provides height agreements, h_p, for samples with forest heights vetted against thresholds, d_n, in Section 4:

$$h_p = \{\, (F_h^k, y_n) \mid y_n \in Y_n, F_h^k \cap h_n\ , if\ height\ thresholds, h_n, are\ 'true'\ estimates\}; \quad (5)$$

$$h_n = \lim_{F_h^k} \left(\min_d h_n \to \max_d h_n\right) \vee F_h^k \gtreqless d_n. \quad (6)$$

Equations 5-6 ratify the accuracy of label agreements using forest height thresholds. These steps allow the rectification of user-defined label agreements to circumvent interpretation biases (Stehman, 1999). Also, four levels were rendered informed by a sample's cumulative score (L0 being the least reliable and L3 the most reliable). We selected L3 reference samples with ≥500 m distributions to avoid issues of spatial autocorrelation (Zandler et al., 2022). The remaining L3 samples were merged with L2 thematic agreements to derive a metrics dataset (Table 2).

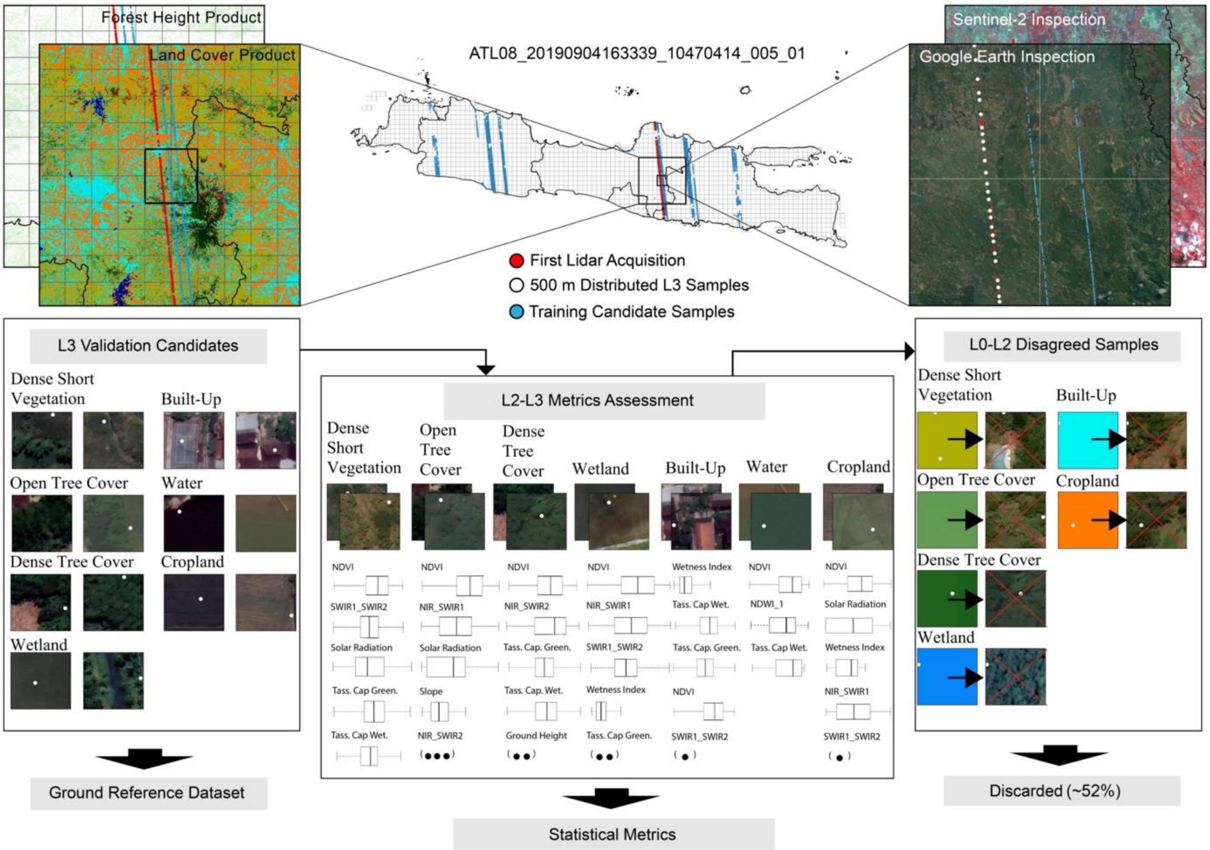

Fig. 3. Lidar data distributions and level allocations. Training samples extend over wide level 1 administrative catchment areas. Pixels correspond to the first lidar acquisition and features are sorted by level of importance with unseen features represented by black dots. **(2 columns) (B/W)**

	Samples	L0	L1	L2	L3
First Lidar Acquisition	4,411	279	1,094	1,801	1,237
	[2.67]	(1.186)	(1.055)	(0.426)	(0.004)
	100%	6%	25%	41%	28%
Metrics Dataset	1,888	-	-	896	992
	[0.43]			(0.426)	(0.004)
	43%	-	-	47%	53%
Filtered Units	2,278	279	1,094	905	
	[2.67]	(1.186)	(1.055)	(0.426)	-
	52%	12%	48%	40%	-
Reference Units (500 m)	245				245
	[0.00]	-	-	-	(0.004)
	6%	-	-	-	100%

Table 2. Sample allocations. **(1 column)**

Note: The 0-1 loss decomposition of total error in parenthesis is estimated using native and vetted labels. Cumulative total errors in brackets quantify the efficiency of label supervisions. For example, the metrics dataset has 83.9% of the cumulative error removed. However, reference units were not measured against a higher level of ground truth and therefore cannot represent a true estimate of error in this table.

The model utilized metric samples to downstream 2019 Sentinel-2 signals from GEE. The year 2019 considers lidar acquisition dates (Table 1), near cloud-free (<10%) images, and the optical throughput of the ICESat-2 receiver expected to work “as designed” over the first nine months (09/2018-06/2019) of the mission (Neumann et al., 2019). Signal extraction was defined by periods of vegetation growth in Indonesia including the withdrawal of wet season period (March-April), a peak growing period (May-June), and the senescence period (July-September) (Apriyana et al., 2021). Multitemporal features were then interpolated with the static morphological features in Table 1 to derive a descriptive set of multidimensional samples.

4.2 Sample supervisions

4.2.1 Metrics assessment

In this section we retrieved amplitude metrics to characterize regional land cover classes. The spectral magnitude of raw signals was smoothed with a moving-average window of 101 indices to retain noise-free temporal profiles. Cropland NDVI, blue, near-infrared, and infrared indices were processed separately to extract the 25^{th}, 50^{th}, and 75^{th} quartiles. Additionally, the mean temporal signal of each metric sample was estimated as shown in our Supplementary Information (Fig. S1). Then, we performed multinomial regressions with the following dependent variables: forest heights (treed classes), SAVI (wetland and cropland), slope (water), and BN (built-up). Finally, we identified statistically significant features (p-value ≥90%) to extract relative bounds (± standard error) of smoothed, mean, and raw vectors (Table S1).

4.2.2 Detection of reliable training samples

Candidate samples were retrieved from eighteen ICESat-2 ground-track acquisitions to meet the training quota. Then, we removed no data values, noisy data, and points landing ≥45 m away from inter-class perimeters. The prevailing samples were concatenated with canopy heights, forest heights, vegetation indices, and DEM-derived features to be used as data input.

A filtering tool was parsed with multitemporal raw metrics given the quantity of thematic agreements in Table S2. The tool queries h_mean_canopy as the primary ICESat-2 value, or queries h_canopy_20 as the secondary value to infill no data segments and return a densely populated canopy height column. Discrete samples are vetted by 3-8 metric bounds, F_n, to find whether the signal of a sample s with class i is within the spectrum of that class: $\min_i F_n \leq s \leq \max_i F_n$. Agreement is given to samples lying within all metric bounds. Then, correlated forest and canopy heights are granted with structural agreements as in sub-section 4.1. Lastly, class-specific forest heights (treed classes) or ICESat-2 canopy heights (non-treed classes) are queried to seek height agreements with equations 5-6. The tool adds the cumulative score of each sample and provides a compact dataset with classed files for user-based discriminations.

4.2.3 Training data refinements

Refinements to a pool of 167,805 samples were carried out to alleviate label uncertainties (Table 3). In total, 37% L3, 41% L2, 20% L1, and 2% L0 sample units were detected. We applied the following logic to eliminate samples of dubious quality:

1. L0: Zero points were attributed to 4,004 samples. Labels failing to pass at least one filter were immediately erased.
2. L1: One point was given to 34,053 samples. Labels with thematic agreement, but unreliable heights were considered liable. Also, height agreements with no thematic agreement were deemed to retail loss. Therefore, L1 samples were all eliminated.
3. L2: Two points were assigned to 68,365 samples. They had (i) thematic and height agreements, (ii) thematic and structural agreements, or (iii) height and structural agreements. We erased all thematic disagreements, except for rare classes.

4. L3: Three points were granted to 61,383 samples considered of the highest quality.

The elimination of 64,702 samples rendered 103,103 reliable samples for model calibration (details in Table S3). As a result, L3 samples expanded to 60% of the training set with cropland samples stratified into cultivation (22%), senescence (37%), and peak growing (41%) classes having 11%, 15%, and 13% L3 apportions, respectively.

	Samples	L0	L1	L2	L3
Pre-	167,805	4,004	34,053	68,365	61,383
Filtering	[2.04]	(0.606)	(0.545)	(0.513)	(0.374)
	100%	2%	20%	41%	37%
Post-	103,103			41,720	61,383
Filtering	[0.89]	-	-	(0.513)	(0.374)
	61%	-	-	25%	37%
Filtered	64,702	4,004	34,053	26,645	
	[1.66]	(0.606)	(0.545)	(0.513)	-
	38%	2%	20%	16%	-

Table 3. Data calibration summary. **(1 column)**

Note: The 0-1 loss decomposition of total errors in parenthesis was estimated using preliminary predictions (sub-section 4.6.1) as the y-variable and native samples (land cover product) as the x-variable. Brackets depict the cumulative loss in error. Refined samples saw 56.4% of the cumulative error removed. Replacing the remaining L2 with L3 samples would reduce 81.7% of the error.

4.3 Model optimization

Multidimensional datasets integrate features to generate predictive power but using unreliable test data underestimates model scores (Burke et al., 2021). Here, the model takes advantage by discriminating contending features based on the mean decrease in impurity (MDI) (Pedregosa et al., 2011). This method performs well with tree-based models, such as Random Forest classifiers, where the impurity is quantified by splitting the criterion of decision trees. Even though Random Forest classifiers have been shown to withstand noise and small training sets (Maxwell et al., 2018), MDI can convey up-ranked features when a model is overfitting. Therefore, permutation feature importance’s were tested over seen and unseen data to find overfitted features if any. This method inspects non-linear estimators by signaling a decrease in model score when a feature is shuffled (Breiman, 2001). However, when two or more features are correlated, equal values are given, and collinearity can be identified. Therefore, we found ten viable combinations shown in Table S4.

SHAP value estimations followed to determine which features expedite the best relative contribution to the detection of land use classes. SHAP values compute explanatory predictions from black box learning models using ideas from game theory (e.g., Li, 2022). They are sensitive to correlated predictors, but the trials above eliminated potential doubts. Fast TreeSHAP (Yang, 2021) was employed taking advantage of the localized predictive accuracy of SHAP values to secure sensitive land use features from Fig. 4.

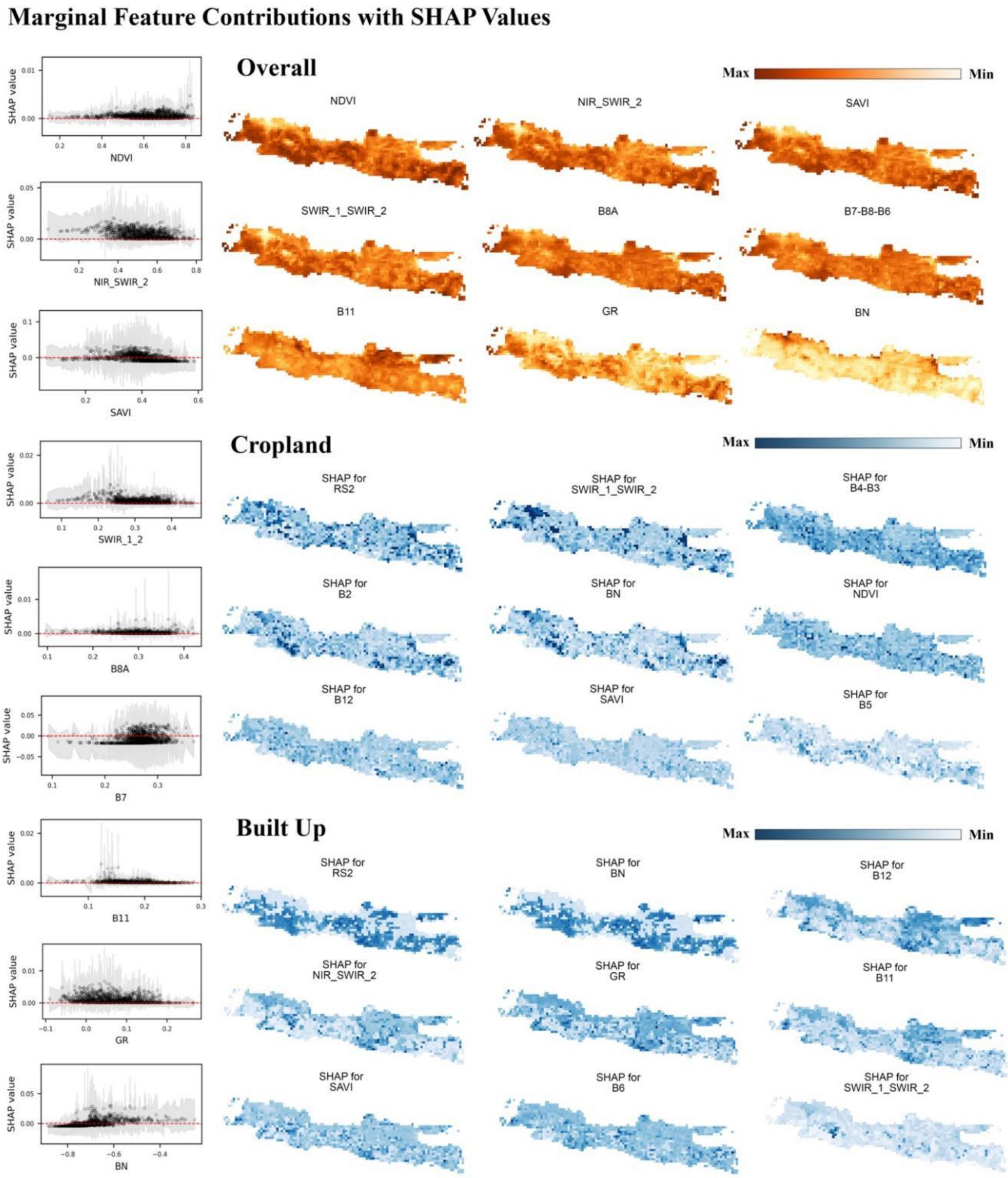

Fig. 4. Feature contributions with SHAP values. Partial dependence scatter plots show overall SHAP values with shaded bootstrap 95% confidence intervals, sorted from highest to lowest. **(2 columns) (B/W)**

4.4 Parameters

Candidate feature combinations were optimized with red (RS2), green (GR), and blue (BN) band indices to avail synergistic performances. However, learning models require unique settings to apply their learnt capacity. Therefore, we sought parameters using a cross-validated grid search approach with five K-fold splits per cross-validated score to avoid error insertion. This was mediated through a stratified shuffle split algorithm of 100 bootstrapped sets of 1000 samples each. Final scores were based on the predictive mean accuracy of trained classifiers over unseen training data. Table 4 shows model performances with error terms.

Pre-Selected Models

No.	*Features*	*(mtry, depth, min.samp.split)*	*(#feat, max_feat.)*	*max samples*	*RMSE*	*Test Data Accuracy*	*Bias-squared*	*Variance*	*Total error*
Model 1	[N, N2, S12, TCG, RS2, GR, S]	(391, 17, 23)	(18, log2)	0.8	3.12	0.78 (0.005)	7.58	0.64	8.22
Model 2	[N, NDWI-2, NS2, S12, TCW, RS2, S]	(391, 19, 15)	(18, log2)	0.8	3.19	0.73 (0.005)	7.77	0.78	8.55
Model 3	[N, NS2, S12,TCW, RS2, S, BN]	(391, 19, 16)	(18, log2)	0.8	3.21	0.73 (0.005)	7.87	0.61	8.48
Model 4	[N, NDWI 2, NS2, S12, RS2, BN, I]	(391, 19, 15)	(18, log2)	0.8	3.19	0.73 (0.004)	7.97	0.56	8.53
Model 5	[N, NS2, S12, TCW, RS2, BN, I]	(391, 17, 23)	(18, log2)	0.8	3.07	0.78 (0.003)	7.94	0.75	8.69
Model 6	[N, NDWI 1, S12, TCW, RS2, GR, BN]	(391, 17, 19)	(18, log2)	0.8	3.31	0.78 (0.004)	7.68	0.56	8.24
Model 7	[N, NS2, S12, RS2, GR, SAVI, I]	(391, 17, 22)	(18, log2)	0.8	3.37	0.84 (0.002)	7.44	0.49	7.93
Model 8	[N, NDWI 1, S12, TCW, RS2, BN, I]	(390, 17, 25)	(18, log2)	0.8	3.37	0.79 (0.003)	7.81	0.63	8.44
Model 9	[N, NS2, S12, RS2, GR, BN, I]	(391, 17, 20)	(18, log2)	0.8	3.35	0.78 (0.003)	8.07	0.66	8.73
Model 10	[N, NS2, S12, RS2, GR, S, BN]	(391, 17, 22)	(18, log2)	0.8	3.08	0.78 (0.005)	8.11	0.61	8.72

Table 4. Comparison of Random Forest models with 95th percentile standard errors in parenthesis. The bias, variance, and total error of predictive accuracies are included. (2 columns)

Note: Sentinel-2 bands B1, B2, B3, B4, B5, B6, B8A, B11, B12, and three full-zero dummy variables were applied to boost the highest ranking features. N= NDVI, S= SAVI, and I= IRECI.

4.5 Bias, variance decomposition

In this section, we decompose the bias and variance of each model's predictive accuracy to estimate the reliability of L3 reference samples. Pixel-based accuracy assessments over preliminary classifications of the 2019 east Java composite were compared using a supervised mean-weighed (SMW) sampling approach and a Stratified Random Sampling (SRS) design (Table A5). SRS includes 269 stratified samples in area-class proportions with a buffered area of ≥500 m. Pixel landings were photo-interpreted using VHR imagery (2019-2021) in Google Earth. The reliability of SMW, x_i, and SRS, y_j, were evaluated using the Mean Absolute Difference (MAD) of producer's (10.55%) and overall (9.07%) accuracies listed in Table S6.

$$\mathrm{MAD} = \frac{1}{n^2}\sum_{i=1}^{n}\sum_{j=1}^{n}\left|x_i - y_j\right| \quad (7)$$

Equation 7 describes a sample size n for a distributed population of absolute sample values j_i, where $i = 1$ to n estimated as the average of all possible absolute differences.

Results of the MAD led us to estimate the loss in accuracy with an expectation (E) over the training set (not individual samples), subdivided into bias and variance following the 0-1 loss decomposition (Dietterich and Kong, 1995). The loss function states that models not predicting true labels are assigned with a bias value of 1, or otherwise 0.

$$\mathrm{Bias} = \begin{cases} 1 \text{ if } y \neq E[\hat{y}], \\ 0 \text{ otherwise.} \end{cases} \quad (8)$$

$$\mathrm{Var} = P(\hat{y} \neq E[\hat{y}]) \quad (9)$$

Equation 8 states that main predictions not always agree with the true label y, while the variance in equation 9 defines the probability, P, that a predicted label does not match the main prediction. However, if the bias is 1, then the loss is entirely defined by the variance; $\{1 - P(\hat{y} \neq E[\hat{y}])\}$. This helps to understand that bias and variance are inversely proportional. So, if a classifier has a high bias causing the main prediction to be always wrong, increasing the variance can be beneficial (Domingos, 2000). We used this function to estimate the predictive error of each model using SRS and SMW reference labels (Table 5).

	Accuracy		*Loss*		*Bias*		*Variance*		*Total Error*	
	SRS	SMW	SRS	SMW	SRS	SMW	SRS	SMW	SRS	SMW
Model 1	0.649	0.739	0.335	0.278	0.333	0.278	0.064	0.009	0.397	0.287
Model 2	0.638	0.746	0.343	0.229	0.333	0.222	0.073	0.026	0.406	0.248
Model 3	0.657	0.746	0.306	0.213	0.315	0.204	0.109	0.033	0.424	0.237
Model 4	0.627	0.705	0.366	0.263	0.352	0.259	0.108	0.024	0.46	0.283
Model 5	0.66	0.728	0.305	0.21	0.296	0.204	0.048	0.03	0.344	0.234
Model 6	0.657	0.75	0.361	0.241	0.37	0.222	0.028	0.069	0.398	0.291
Model 7	0.66	0.75	0.323	0.259	0.315	0.259	0.055	0.02	0.37	0.279
Model 8	0.657	0.754	0.315	0.223	0.315	0.222	0.036	0.051	0.351	0.273

Model 9	0.657	0.743	0.335	0.223	0.333	0.222	0.066	0.024	0.399	0.246
Model 10	0.66	0.769	0.325	0.219	0.315	0.204	0.054	0.039	0.369	0.243

Table 5. Error quantification with two reference sample designs. **(2 columns)**

Note: The ten Random Forest classifiers in Table 4 are learnt with the same data. This table compares the utility of photo-interpreted and bias-reduced reference samples and measures the error inherent to interpretation bias.

All models show that SRS labels inherit a higher bias and variance. We also estimated the mean total error for SRS (0.39) and SMW (0.26) labels including the structural (0.64), thematic (0.56), and height (0.54) agreement mechanisms used. The assessment revealed that L3 reference samples had a 33% smaller mean error (Table S7). So, labeling with VHR imagery and vetting those labels with auxiliary information serves to increase confidence over the reported metrics of change (Olofsson et al., 2014).

4.6 Classifications and accuracy assessments

4.6.1 Preliminary classifications

The model with highest predictive accuracy was selected to learn unbalanced samples proportional to area-class pixel landings. SMW and SRS preliminary accuracy assessments were conducted to verify whether the variance of user's accuracies reflected the results in sub-section 4.5. Confusion matrixes were resolved with AcATaMA (Llano, 2022), a QGIS plugin to estimate map accuracy expressed in area-proportions. Table S8 (SMW) and Table S9 (SRS) confirmed the utility of L3 reference labels with large sample sizes.

4.6.2 Informed classifications

Final-stage post-processing steps are reported here with parameters, sequences, and random seeds to remove erroneous pixels and improve the temporal consistency of preliminary classifications (Li et al., 2022). We applied a morphological processing sequence starting with a dilation operation over rare classes (water and wetlands) with 1-pixel size to shape domain perimeters and avoid pixel intrusions. Pixels smaller than a two-pixel threshold with 0-pixel connections and a random seed of 0 were then removed. A second dilation was performed over target classes (open tree cover, dense tree cover, built-up, and cropland) with 1-pixel size to consolidate their perimeters. Temporal co-registrations with the 2019 template concluded the sequence and increased the overall accuracy from 78% to 82% (Table S10).

4.6.3 Post-informed classifications

Post-informed bi-temporal maps were produced followed by accuracy assessments of the 2017-2019 east composite of Java. Maps account for seven strata classes and five transition classes shown in Table 6. We also sought new L3 reference labels with power allocations over rare and transition classes to ensure lower variance in the error (Cochran, 1977). All reference samples were reviewed to detect class transitions with the time-stamp tool offered by Google Earth. We used a 3 x 3-pixel kernel window to ensure that 50% of the neighboring pixels went through the same change as the central pixel. Also, trends before and beyond specific target years were revised to rely on transitions that were not a product of seasonal or annual weather change. The post-informed reference set reflects 'from-to' labels for transition classes representing mean-weighed stratifications and power allocations shown in Table S11.

Reference	*Dense Short Vegetation*	*Open Tree Cover*	*Dense Tree Cover*	*Wetland*	*Built Up*	*Water*	*Cropland*	*Built Up Loss*	*Built Up Gain*	*Cropland Loss*	*Cropland Gain*	*Cropland > Built Up*	*Built Up > Cropland*	*User's Accuracy*		*Error-Adjusted Area (ha)*	
	1	2	3	4	5	6	7	8	9	10	11	12	13	14		15	
1	1.49%	0.45%	0.15%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.15%	0.00%	0.00%	0.00%	66.67%	± 23.9%	318,916	± 21.2%
2	0.00%	8.41%	0.49%	0.00%	0.32%	0.00%	0.81%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	83.87%	± 9.2%	1,146,244	± 9.1%
3	0.22%	0.67%	12.34%	0.00%	0.22%	0.00%	0.90%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	85.94%	± 8.5%	1,687,818	± 9.8%
4	0.00%	0.00%	0.00%	1.02%	0.00%	0.51%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	66.67%	± 53.3%	359,167	± 48.2%
5	0.00%	0.00%	0.00%	0.00%	8.08%	0.21%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	97.44%	± 5.0%	971,793	± 7.5%
6	0.00%	0.00%	1.26%	2.53%	0.00%	17.70%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	82.35%	± 18.1%	1,752,631	± 11.5%
7	1.35%	1.35%	0.68%	0.23%	0.45%	0.00%	23.22%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	85.12%	± 6.3%	2,710,664	± 4.6%
8	0.00%	0.00%	0.00%	0.00%	0.37%	0.00%	0.37%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	± 0.0%	0	± 0%
9	0.00%	0.00%	0.00%	0.00%	0.16%	0.00%	0.16%	0.00%	0.47%	0.00%	0.00%	0.00%	0.00%	60.00%	± 42.9%	45,069	± 40.8%
10	0.00%	1.01%	2.52%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	1.51%	0.00%	0.00%	0.00%	30.00%	± 28.4%	158,101	± 47.2%
11	0.00%	0.00%	0.30%	0.00%	0.61%	0.00%	0.61%	0.00%	0.00%	0.00%	2.43%	0.00%	0.00%	61.54%	± 26.4%	231,001	± 22.8%
12	0.00%	0.16%	0.00%	0.00%	0.00%	0.00%	1.29%	0.00%	0.00%	0.00%	0.00%	0.81%	0.00%	35.71%	± 25.1%	76,830	± 37.2%
13	0.29%	0.00%	0.00%	0.00%	0.00%	0.00%	1.14%	0.00%	0.00%	0.00%	0.00%	0.00%	0.57%	28.57%	± 33.5%	54,195	± 64.5%
Producer's Accuracy	44.46% ± 6.8%	69.77% ± 6.5%	69.57% ± 7.0%	27.06% ± 11.6%	79.09% ± 7.2%	96.07% ± 17.3%	81.49% ± 5.8%	0.00% ± 0.0	100.00% ± 12.2%	91.03% ± 24.4%	100.00% ± 13.2%	100.00% ± 14.5%	100.00% ± 27.3%	OA = 78.06 ± 5.06%			

Table 6. Confusion matrix of post-informed land cover change with error-adjusted areas and classes estimated with area proportions shown in percentages with 95% confidence intervals. **(2 columns)**

Note: User's confidence intervals (14) have fewer variances including the error margin of overall accuracy (OA) in contrast to the confusion matrix of informed land cover classification (Table S10).

5. Evaluation

5.1 Mapping outcomes

Bi-temporal land change maps (2016-2021) were produced for the geographical unit of Java. The reporting scheme analyzes open tree cover and dense tree cover as a merged forest class. However, these classes retain their original stratification on the map. Further, stable strata were used as a base layer pertaining to the latter year of change instead of the former to contrast a target year’s ecosystem. Results are reported in percentages rather than absolute values to compare relative changes and map names use the ISO 3166 country code, a three-letter code (alpha-3) followed by temporal periods and unit coordinates (e.g., [IDN]2017201910S110E.tif).

5.2 Analysis zones

The evaluation selects two non-affected baseline areas and seven resettlement sites shown in Fig. 5. Site selections considered size of settlement and overlapping areas of influence out of the 23 resettlement sites in the Cangkringan district. We also paid particular attention to the main roads near resettlement sites given their capacity to propagate intensive land use changes. Areas of influence were drawn from the centroid of each settlement to represent the assumed daily median walking distance, equivalent to 800 m (Yang and Diez-Roux, 2012).

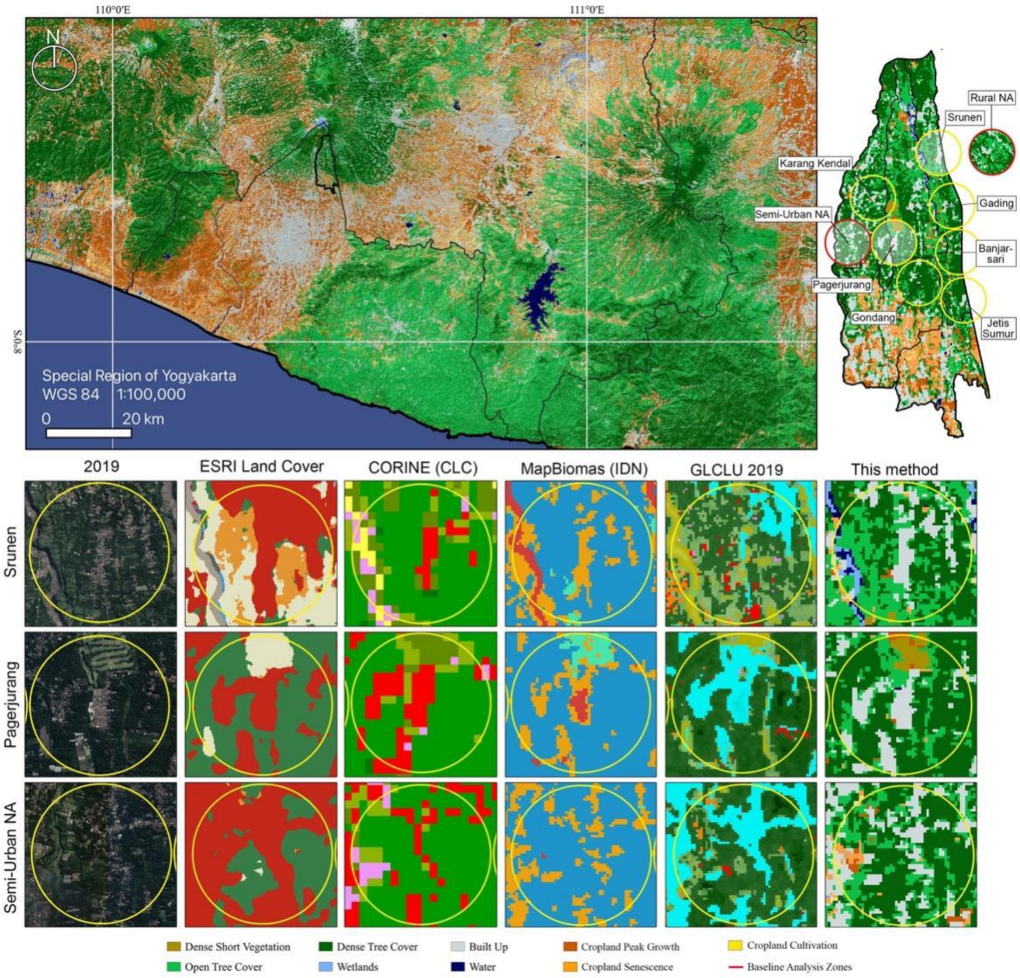

Fig. 5. Classification of the Special Region of Yogyakarta in 2019 with Mt. Merapi located in the northern apex. The top right is the Cangkringan district with seven affected (AF) and two non-affected (NA) zones. The first column denotes 2019 Google Earth images and the next set of columns show global land cover maps sorted by relative precision. Rows depict rural (Srunen) and semi-urban (Pagerjurang) resettlement sites with a non-affected (baseline) area at the bottom. **(2 columns) (B/W)**

Land change interactions were sought by comparing sub-district and localized trends to extract the effects of urban resettlement in rural areas. Comparative change distributions, relative intensities of change, and livelihoods in terms of crop area change determine the extent of these trends. The first interaction was identified at the district level in Fig. 6, where rural areas show continuous built-up growth caused by forest loss.

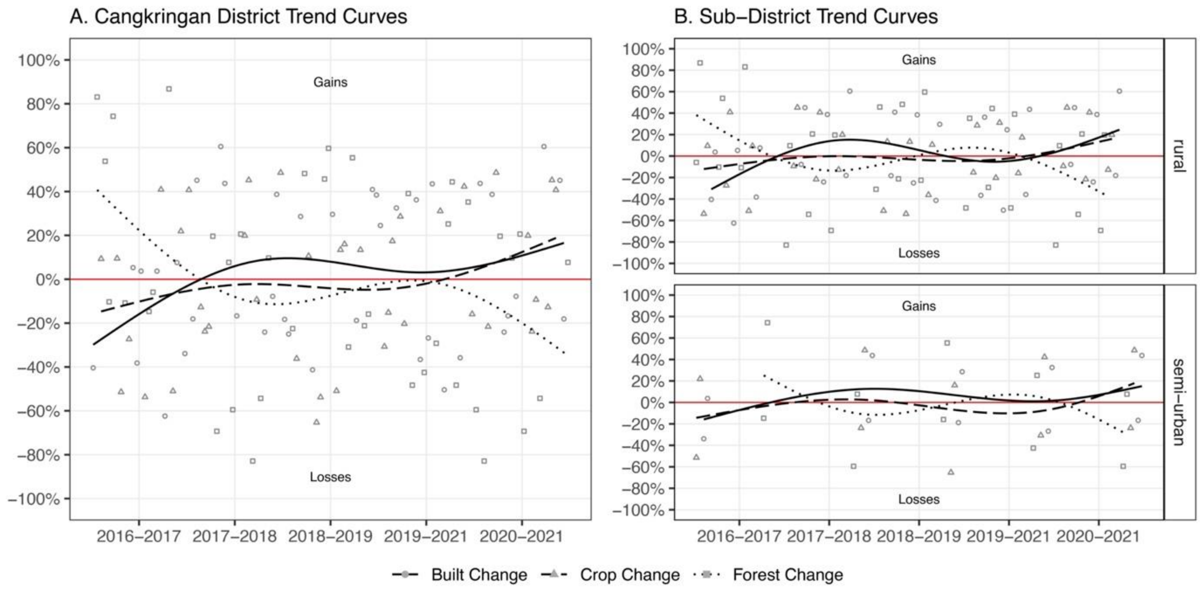

Fig. 6. Land changes in the Cangkringan district. Values are expressed in percentages of total change. Smoothed trend lines use Gaussian regression with four-degree polynomial curves. **(2 columns) (B/W)**

5.3 Sub-districts' land interactions

Intra-annual seasons s of vegetation growth (growing, peak, and senescence) were extrapolated from cloud-free Sentinel-2 satellite images to gauge inter-seasonal net cropland mean changes (Table 7). We first collected bi-annual area changes, Δ_{y_i}, between 2017, 2018, 2020, and 2021 subdivided into gains, g, and losses, l, while avoiding other land cover types:

$$f_g \ = \ \{(g,s) \mid g \in s \ \forall \ y_i \ , \textstyle\sum_{i>1}^{y}(\Delta \ y_i)\}; \tag{10}$$

$$f_l \ = \ \{(l,s) \mid l \in s \ \forall \ y_i \ , \textstyle\sum_{i>1}^{y}(\Delta \ y_i)\}. \tag{11}$$

Equations 10-11 define inter-seasonal gross changes to find the average net change per unit area, $\sum_{y>1}^{\Delta y_i} \left(k_g - k_l\right)/x_s$, further normalized by the number of seasons, x_s. The resulting net changes were confirmed by estimating inter-annual mean changes:

$$k_g \ = \ \{(g,s) \mid g \in \Delta \ s_i \ \forall \ s_i, \textstyle\sum_{i>1}(\Delta \ s_i)/x_s\}; \tag{12}$$

$$k_l \ = \ \{(l,s) \mid l \in \Delta \ s_i \ \forall \ s_i, \textstyle\sum_{i>1}(\Delta \ s_i)/x_s\}; \tag{13}$$

where equations 12-13 derive an inter-seasonal average of bi-annual changes followed by the net change estimate of unit areas, $\sum_{y>1}^{\Delta y_i}\left(k_g - k_l\right)$. We also analyzed 2016-2021 mean land use transitions to illustrate the overall geographic distribution of post-disaster changes and the standard deviation of transition counts as shown in Fig. 7.

	Inter-seasonal change				Inter-annual change			
Affected sub-districts	2017 2018	2018 2020	2020 2021	Net Change	Month 04-05	Month 06-07	Month 08-09	Mean Change
Glagaharjo	-23.0	-17.6	11.7	-28.9	-49.0	-7.90	-29.7	-28.9
Kepuharjo	-41.8	-14.6	21.9	-34.5	-54.4	72.3	-121.3	-34.5
Umbulharjo	-51.8	-8.1	45.3	-14.6	-23.0	115.0	-136.0	-14.6
Total Affected	-116.6	-40.3	78.9	-78.0	-126.1	179.2	-287.0	-78.0
Total Non-Affected	27.8	-80.9	72.4	19.3	114.8	181.8	-238.4	19.3

Table 7. Intra-annual seasons of cropland area change (Ha) **(1 column)**

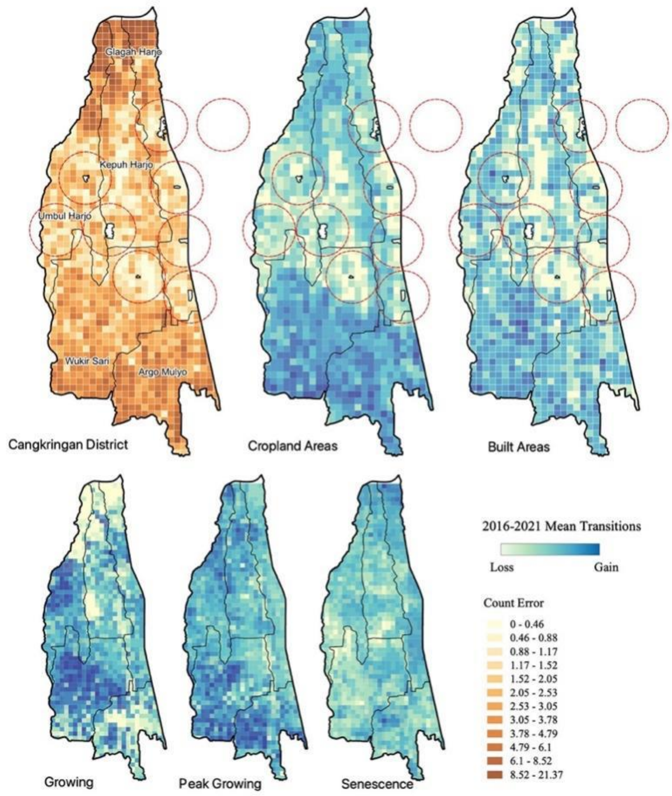

Fig. 7. Mean transition counts. 200-m tiles estimate the mean value of transition counts with higher gain intensities in northern areas of the affected sub-districs (Umbulharjo, Kepuharjo, and Glaga Harjo) and southern areas of the unaffected sub-districts (Wukirsari and Argo Mulyo). Count errors are shown with class intervals by standard deviation from the mean error, which is zero. **(1 column) (B/W)**

The affected areas denote equivalent inter-seasonal and inter-annual cropland net changes with a high amount of cropland losses. Non-affected areas slightly grew during the analysis period. However, the sub-districts have several topological and topographical variations, thus required a contrastive analysis against localized areas of influence.

5.4 Resettlement areas’ land interactions

Land use and land cover interactions of the analysis period now focus on geographically homogeneous locations. Table S12 provides land change measures between affected and non-affected areas of influence. We saw that land use exchange experienced low intensity changes near semi-urban areas, while built gain peaked in rural areas. Also, non-affected areas had significantly higher cumulative changes. This can be explained by the intensity of peri-urban dynamics increasing since before 2010 (Giyarsih, 2010). Another observation is the intensity of forest to built-up change growing from Gading (rural) to Karang Kendal (semi-urban). Fig. 8 illustrates these trends dividing rural and semi-urban resettlement areas.

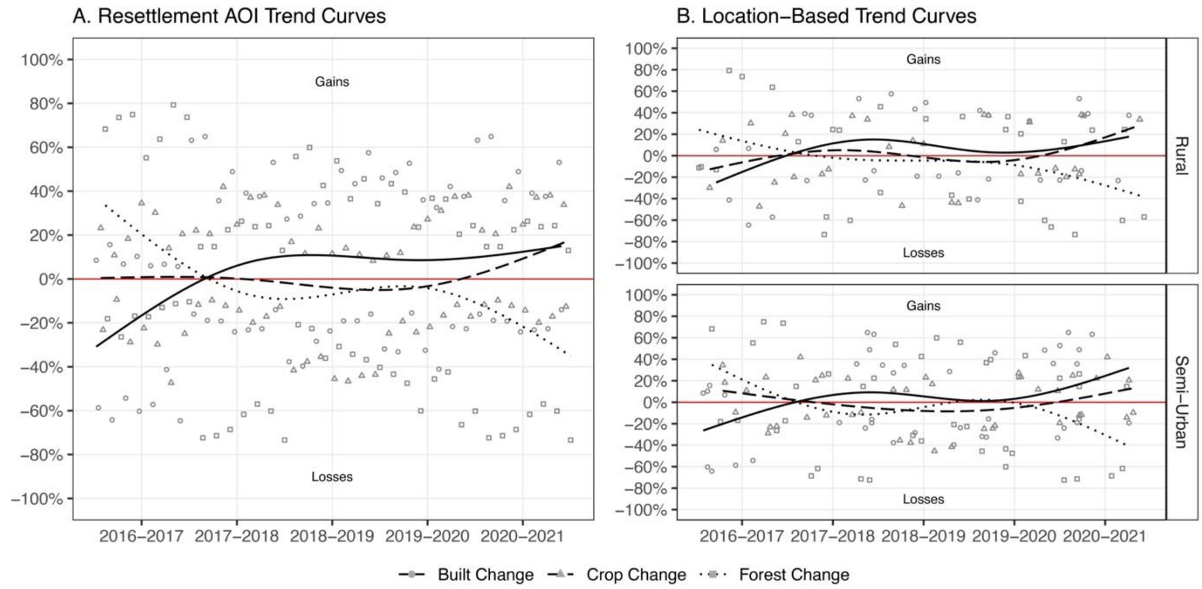

Fig. 8. Land changes in resettlement areas of influence (AOI). Smoothed trend lines use Gaussian regression defined by four-degree polynomials. **(2 columns) (B/W)**

The analysis saw a steep increment of built land in 2018 increasing until 2021 with an inversely proportional change in forest area. This is assumed to be a long-term effect of resettlement (2014-2018), further exacerbated by Yogyakarta’s spillover effects (Widowaty and Oktavian Artanto, 2018). Meanwhile, semi-urban areas depict moderate cropland losses.

We also analyzed peri-urban dynamics comparing 2016-2021 mean change rates in affected sub-districts and affected resettlement areas. Table 8 shows relative gross and net change intensities sorted from rural to semi-urban areas. Forests represent the highest net change output values for sub-districts (4.5%) and resettlement areas (-6.9%). Conversely, built areas manifest an increase of 0.3% on the aggregate level with localized areas growing at a mean rate of 4.7%. In particular, cropland growth in resettlement areas show that spillover effects were successfully undermined given the overall declining tendency in the region.

		Crop Gain	*Crop Loss*	Net Change	*Built Gain*	*Built Loss*	*Net Change*	*Forest Gain*	*Forest Loss*	*Net Change*
Sub-District	Glagaharjo	14.3%	30.5%	-16.2%	34.4%	31.4%	3.0%	51.3%	38.1%	13.2%
Rural	Srunen	26.6%	22.0%	4.6%	38.1%	29.8%	8.3%	35.3%	48.2%	-12.9%
Rural	Gading	29.4%	27.4%	2.0%	31.7%	34.2%	-2.5%	38.9%	38.4%	0.5%
Rural	Banjarsari	28.9%	26.4%	2.5%	33.6%	25.4%	8.2%	37.5%	48.3%	-10.8%
S-Urban	Jetis Sumur	31.4%	21.1%	10.3%	31.8%	27.5%	4.3%	36.8%	51.3%	-14.5%
Sub-District	Kepuharjo	33.0%	29.9%	3.1%	28.5%	37.7%	-9.1%	38.5%	32.5%	6.0%
S-Urban	Pagerjurang	20.0%	22.3%	-2.3%	37.8%	35.0%	2.8%	42.3%	42.7%	-0.4%
Sub-District	Umbulharjo	28.0%	29.4%	-1.4%	30.0%	22.8%	7.1%	42.0%	47.8%	-5.8%
S-Urban	Karang Kendal	15.4%	17.2%	-1.8%	45.5%	33.9%	11.6%	39.1%	48.9%	-9.8%
S-Urban	Gondang	24.6%	24.4%	0.3%	34.2%	34.2%	0.0%	41.1%	41.4%	-0.3%
Mean	Sub-Districts	25.1%	29.9%	-4.8%	31.0%	30.6%	0.3%	43.9%	39.5%	4.5%
Total	Settlements	25.2%	23.0%	2.2%	36.1%	31.4%	4.7%	38.7%	45.6%	-6.9%

Table 8. 2016-2021 Mean intensity of change over total change. **(1 column)**

6. Discussion and concluding remarks

6.1 Automatic sample calibrations

The model described in this study funnels multisource remote sensing data to vet the reliability of land cover labels. A regionally-consistent set of multitemporal metrics were acquired after eliminating 52% dubious samples. These metrics support–via rule-based combination of attributes–multiple classification schemes to suit a range of science communities and policy makers with high-quality training datasets (Gomez et al., 2016). The amplitude of multitemporal signals in Fig. S2 outperformed smoothed amplitudes, both limited to a transect of Java. Smoothed metrics delivered 66,917 reliable samples and a 56% overall mapping accuracy. We also used mean temporal profiles to extend the number of thematic agreements. The model was learnt on 81,924 reliable samples providing a 74% overall accuracy. In light of these results, we selected raw multitemporal metrics to secure a high-fidelity profile of regional amplitude features representing the land cover classes of interest.

6.2 Model optimization and feature performances

Regional or national classifiers are subject to various topological patterns of the Earth challenging model selection processes (Belgiu and Csillik, 2018). Uncorrelated feature candidates ensured predictive efficiency while avoiding to overfit the models. These were evaluated with L3 reference samples and the bias-variance decomposition function to select a final model. The Random Forest classifier showed a considerable sensitivity to the number of training samples per class. Therefore, we aimed to maintain pixel landing proportions by filtering entire levels of training data instead of specific class units.

SHAP values highlighted the utility of localized features for class-oriented accuracies. NDVI was the best feature, while NIR-SWIR2 was highly ranked and not identified using the MDI algorithm. NIR-SWIR1 saw unstable performances across all classes. SWIR1-SWIR2 was particularly useful in identifying land use classes, but Red/SWIR2 and Blue/NIR were found to be the highest contributors to build and crop land (Fig. S3).

6.3 Ground reference samples and accuracy

The loss in error function led to a general understanding that biased labels are a product of human interpretation. This was clarified when comparing the variance of user accuracies in Table S8 and Table S9. The cumulative difference was 9.3% with a mean difference of 18.9% between classes (see Table S13). Therefore, bias-reduced labels narrowed the mean variance of all user accuracies from 17.35% (SRS) to 11.83% (post-informed), thus eliminating 32% of the bias. Consequently, the net worth of producer and user accuracies grew by 8.4% over the land cover product (Hansen et al., 2022), thereby enabling change detection analyses.

6.4 Land interactions in the Cangkringan district

The evaluation of target areas progressively isolated peri-urban dynamics to measure land change effects of resettlement. We also sorted transition counts to find a logical sequence of land change events (Fig. 9). The Cangkringan district saw forest to cropland change (Pearson's r=68-87), then forest to built-up change (Pearson's r=45-62), and followed by cropland to built-up change (Pearson's r=26-52). Therefore, forests are exchanged for cropland as a medium to built-up land. We also noted 19.4 ha of intra-annual cropland change near urban areas (<18.9 km from the city center) and -78 ha in rural areas (<6.46 km from the urban fringe). These results mark a cross-sectional divide between urban and rural functions of the region.

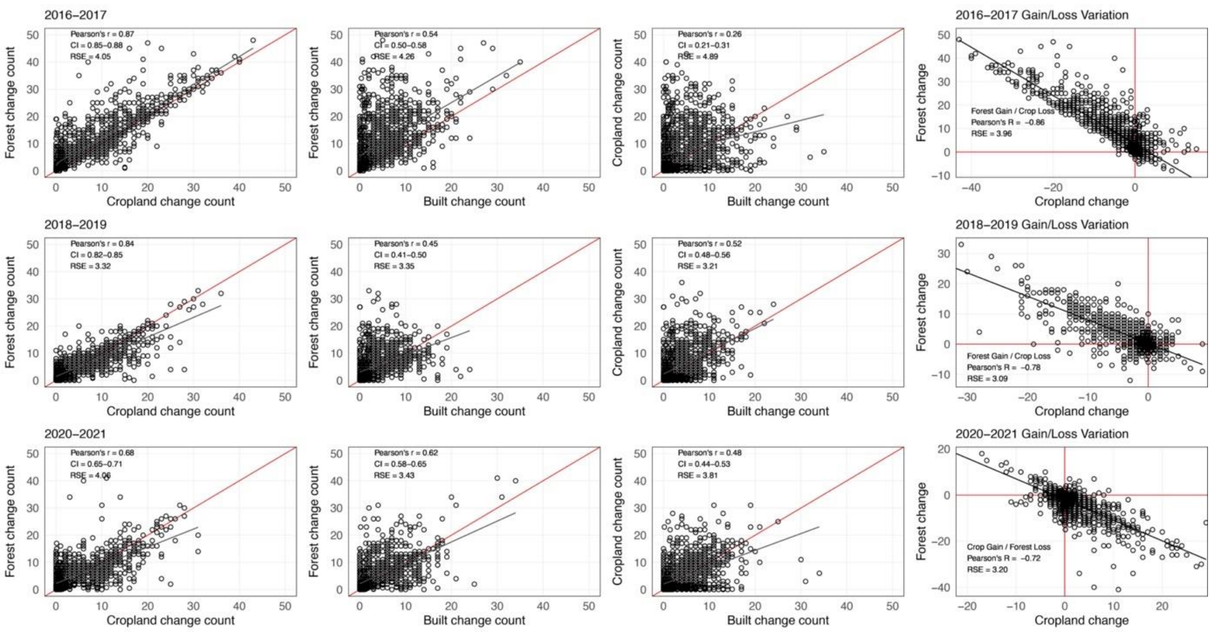

Fig. 9. Correlation of transition counts with best-fit linear regressions for the Cangkringan district. The full series is available in our supplementary material Fig. S4. The right column illustrates gain/loss variations for the highest correlation. Values for Pearson's r, 95% confidence intervals (CI), and residual standard errors (RSE) complement the analysis. **(2 columns) (B/W)**

Localized analyses identified transitions strictly caused by urban resettlement. These areas are enduring intensive forest loss (-6.9%) due to built-up growth (4.7%). Moreover, if peri-urban dynamics, d_i, extend homogeneously over 'Not Affected', x_i, and 'Affected' areas, y_i, the total change of class i in unit areas n_a and a can be defined as:

$$Total_{ch_i} = \sum_i^{n_a} \sum_i^{a} \left\| \left[\frac{x_i}{n_a}\right] + \left[\frac{y_i}{a}\right] \right\| ; \quad (14)$$

$$\mathrm{d_i} = \left\{ \left(\frac{\mathrm{x_i}}{\mathrm{Total_{ch_i}}} \right) \times 100, where\ 'not\ affected'\ areas\ represent\ urban\ dynamics \right\}; \quad (15)$$

$$l_e = \left\{ \left| 1 - \left[\left(\frac{D_i}{100} \right) \times j_i \right] \right|, 'affected'\ areas\ factor\ d_i\ and\ resettlement\ effects \right\}. \quad (16)$$

Equations 14-16 estimate the land change effects, l_e, of investments made. Using land change areas in Table 9, we found that 16.4% forest, 18.5% cropland, and 15.5% built changes were attributed to resettlement sites. Consequently, 7.39 ha of forest loss potentially pertain to efforts of livelihood development. Also, cropland saw 13.97 ± 8.84 ha of growth with a 2.2% mean intensity amid the overall declining tendency (-4.8%). This translates to agricultural development near resettlement sites overturning district-level odds by 1.79 times.

		Gross Cropland Change (ha)			Gross Built Up Change (ha)			Gross Forest Change (ha)		
		Gain	*Loss*	*Net Change*	*Gain*	*Loss*	*Net Change*	*Gain*	*Loss*	*Net change*
1. Rural	Srunen	26.01 ± 1.19	21.54 ± 0.98	4.47 ± 5.46	37.33 ± 2.78	29.21 ± 2.18	8.12 ± 5.2	34.59 ± 3.27	47.19 ± 4.46	-12.60 ± 4.93
2. Rural	Gading	22.73 ± 1.04	21.18 ± 0.97	1.55 ± 5.48	24.46 ± 1.82	26.38 ± 1.97	-1.92 ± 5.31	30.03 ± 2.84	29.66 ± 2.80	0.37 ± 5.13
3. Rural	Banjarsari	21.08 ± 0.96	19.26 ± 0.88	1.83 ± 5.49	24.55 ± 1.83	18.53 ± 1.38	6.02 ± 5.37	27.38 ± 2.59	35.23 ± 3.33	-7.85 ± 5.11
4. Urban	Gondang	15.97 ± 0.73	15.79 ± 0.72	0.18 ± 5.53	22.18 ± 1.65	22.18 ± 1.65	0.00 ± 5.36	26.65 ± 2.52	26.84 ± 2.54	-0.18 ± 5.19
5. Urban	Jetis Sumur	29.94 ± 1.37	20.17 ± 0.92	9.77 ± 5.45	30.39 ± 2.27	26.29 ± 1.96	4.11 ± 5.27	35.14 ± 3.32	49.02 ± 4.63	-13.87 ± 4.9
6. Urban	Pagerjurang	15.52 ± 0.71	17.34 ± 0.79	-1.83 ± 5.52	29.39 ± 2.19	27.20 ± 2.03	2.19 ± 5.27	32.86 ± 3.11	33.22 ± 3.14	-0.37 ± 5.07
7. Urban	Karang K.	16.61 ± 0.76	18.62 ± 0.85	-2.01 ± 5.51	49.20 ± 3.67	36.60 ± 2.73	12.60 ± 5.06	42.26 ± 4.00	52.85 ± 5.00	-10.59 ± 4.8
8. Rural	Not Affected	28.57 ± 1.30	24.55 ± 1.12	4.02 ± 5.44	35.51 ± 2.65	31.13 ± 2.32	4.38 ± 5.2	41.62 ± 3.93	50.02 ± 4.73	-8.40 ± 4.83
9. Urban	Not Affected	29.48 ± 1.35	28.20 ± 1.29	1.28 ± 8.18	51.30 ± 3.82	41.99 ± 3.13	9.31 ± 7.56	55.77 ± 5.27	66.36 ± 6.27	-10.59 ± 6.83
Total Affected		147.87 ± 6.75	133.90 ± 6.11	13.97 ± 8.84	217.51 ± 16.22	186.39 ± 13.90	31.13 ± 11.09	228.92 ± 21.64	274.01 ± 25.90	-45.09 ± 13.93

Table 9. 2016-2021 Land change with area-reference estimators and error-adjusted standard errors expressed in 95% confidence intervals. **(2 columns)**

Note: 95% Confidence intervals for relative net change values consider the sum of squares for gain/loss estimates.

6.5 Implications and limitations

The following points summarize the main implications of this study:

- The GCC model renders a two-step data quality supervision model. First, reliable reference and metric samples are obtained to characterize eco-zones of interest. Second, amplitude metrics are converged in a canopy height vetting system to detect high-quality labeled samples.

- 32% of the variance in user's accuracy was eliminated with bias-reduced reference samples. Also, quality control reduced 56.4% of the loss in error from the training set.
- Rural areas saw 16.4% forest loss, 18.5% cropland growth, and 15.5% built growth as a product of urban resettlement. In addition, the resiliency of recovered livelihoods was quantified with a 2.2% mean net intensity.

Limitations of this study encompass the time-consuming steps to attain statistically significant metrics. The model relies on hand-labeled sample agreements vetted with ancillary datasets to achieve automatic label supervisions at the expense of time. Feature metrics can be enhanced by increasing and dispersing the number of annual temporal acquisitions. Lastly, unreliable samples require a relabeling module for further integration instead of elimination.

6.6 Concluding remarks

This study builds upon the second sub-model of a microsimulation system (Garcia-Fry et al., 2022) to formulate a simple and automatic label supervision model that predicts accurate land cover maps, aiming to evaluate land change interactions prompted by post-disaster resettlement sites. The GCC model produced mapping accuracies of 78%-82% on the first iteration with 8.4% class-improved native accuracies. This enabled site-specific evaluations over rural-affected areas after the 2010 Mt. Merapi eruptions. Empirical results of this study exhibit a post-disaster land change sequence of 16.4% forest loss, 18.5% cropland growth, and 15.5% built-up growth as a product of resettlement. Given Yogyakarta's spillover effects, we believe that rural areas now depend on how future investments accommodate causal growth in compact land use networks. However, policies to consolidate agronomic land in central areas of the network are required. Further, rural fragmentation can be averted by monitoring investments that avail informal land transactions. Tree farming can be used as a hazard mitigation strategy to secure those areas and ensure the sustainability of rural livelihoods.

Future research contends integration of a gridded population mapping algorithm to evaluate the uncertainties associated with post-disaster population dynamics. This study marks a major step towards disaster recovery planning as technological improvements continue to rise with big data models at the forefront of Earth observation.

Author Contributions

Martin Garcia-Fry: Conceptualization, Methodology, Software, Formal analysis, Data curation, Visualization, Investigation, Writing–original draft, Writing–review & editing. **Osamu Murao**: Supervision, Methodology, Visualization, Writing–Review & Editing. **Bruno Adriano**: Formal Analysis, Methodology, Writing–review & Editing. **Shunichi Koshimura**: Formal Analysis, Investigation, Methodology, Writing–review & Editing. All authors read and approved the manuscript.

Declarations of Interest

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